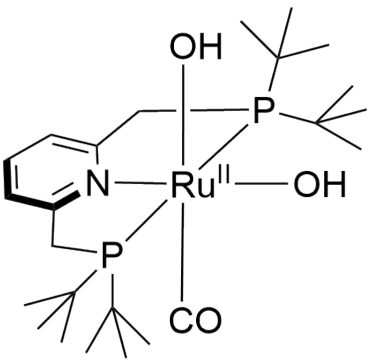
AE-503: Dilution of [Ru(CO)(OH)2(PNP)] solution in water while in-situ IR measurement I

Date: 2025-05-30
Tags: Dihydroxo [Ru(CO)(OH)2(PNP)] AE IR in-situ
Category: Two-photon
Status: Done
Created by: Alexander Eith

Reaction scheme/sample structure



Chemical Formula: $C_{24}H_{45}NO_3P_2Ru$
Molecular Weight: 558,64552

AE-493.cdxml

Literature/reference experiments

Literature	/
Reproduction	/
Similar experiments	Two-photon - AE-473: Testing of in-situ IR probe

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [µmol]	Equivalents	$Mass_{theo}$ [mg]	$Mass_{exp}$ [mg]	Molar mass [g/mol]	Volume [ml]	concentration [mM]
[Ru(CO)(OH)2(PNP)] ([Ru 1])	Two-photon - AE-499: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water	/	/	/	/	/	558.65	0.3	saturated solution
[Ru(CO)(OH)2(PNP)] ([Ru 2])	Two-photon - AE-456: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h	/	5.21	/	/	2.91	558.65	/	/
water	Prep work - AE-442: Degassing of Milli-Q water	/	/	/	/	/	/	0.5 + 0.3	/
MeCN (dry)	AE-152: Degassing of MeCN	/	/	/	/	/	/	0.4 + 1.6	/
MeCN (HPLC grade)	75-05-08	/	/	/	/	/	/	0.5 + 0.8	/

Procedure/observations

All steps were carried out using [Protocol] Schlenk Technique, the flasks were not dried by heating prior to use, unless other specified. All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations	Pictures
		The in-situ IR (Equipment - ReactIR 701L with DST 6.35 x 1.5 m x 305 DiComp probe) was used according to Protocol - In-situ IR measurment with ReactIR 701L in GL14 flask under inert conditions, the steps done are shortly explained in the experiment	/	/
30.05	10:30	The device was started and cooled with liquid nitrogen	/	/
		The following steps were done in air:	/	/
	14:34	The background measurement was performed in air (256 scans)	/	/
	14:36	The Probe was inserted into a GL14 flask 5 mL (flask 1) and secured with the BOLA fitting	/	1 setup flask 1.jpg 1 position of probe in flask 1.jpg
		To flask 1 water (0.5 mL) was added through the valve using a 1 mL syringe	/	/
		From now on stirring: 500 rpm for all samples and flasks during measurment	/	/
	14:37	The measurment was started (sample interval 3 min, experiment duration: 8 h, resolution: 4 cm^-1)	**AE-503-1**	/
	14:55	The measurment was paused	/	/
		From here under inert conditons:	/	/
	14:57	Flask 1 was removed from the probe and replaced with a GL14/NS14 5 mL flask (flask 2), the NS14 was closed with a new rubber septum	/	1 setup flask 2.jpg 1 position of probe in flask 2.jpg
	14:58 - 15:02	flask 2 was sekurated	/	/
	15:04	To flask 2 [Ru 1] was added (0.3 mL)	/	/
	15:05	The measurment was resumed	**AE-503-1**	/
	15:23	The measurment was stopped	/	/
		flask 1 and 2 and the IR probe were cleaned and dried with water, aceton, paper towels and Q tips	/	/

		The following steps were done in air:	/	/
	15:37	The background measurement was performed in air (256 scans)	/	/
		The Probe was inserted into flask 1 and secured with the BOLA fitting	/	2 setup flask 1.jpg
	15:43	To flask 1 MeCN (HPLC grade) (0.5 mL) was added through the valve using a 1 mL syringe	/	/
		The measurment was started (sample interval 3 min, experiment duration: 8 h, resolution: 4 cm^-1)	**AE-503-2**	/
		From here under inert conditons:	/	/
		Flask 2 was set under argon	/	/
	15:45	To flask 2 [Ru 2] was added (2.91 mg)	/	/
	15:57	The measurment was paused	/	/
	15:58	Flask 1 was removed from the probe and replaced with flask 2 (The flask was not sekurated beforehand and the assambly was not done in argon counterflow	/	/
	16:00	flask 2 was set under argon	/	2 setup during sekurating.jpg 2 positon of probe in flask 2.jpg
		The NS14 was changed to a new rubber septum	/	/
	16:09	To flask 2 a full 1 mL syringe with MeCN (dry, 1 mL) was connected (no liquid was added into the flask, it remianed in the syringe)	/	/
	16:10	To flask 2 MeCN (0.4 mL) was added from the syringe	**AE-503-2** dark yellow / orange solution, maybe small amount of solid remains	2 after addition of 0.4 mL MeCN.jpg
	16:11	The measurment was resumed	/	/
	16:23	MeCN (0.2 mL) was added from the syringe	9 measurments collected before addition slightly lighter in colour, definetly completly dissolved	2 after addition of 0.6 mL MeCN.jpg
	16:28	MeCN (0.2 mL) was added from the syringe	11 measurments collected before addition	/

	16:37	MeCN (0.2 mL) was added from the syringe	14 measurments collected before addition lighter in colour, no other change	2 after addition of 1 mL MeCN.jpg
	16:44	The now empty syringe was removed and a new 1 mL syringe filled with MeCN (dry, 1 mL) was added	/	/
		The volume of the syringe was adjusted to excactly 1 mL (pushing out the remaining air bubble)	16 measurments collected before adjustment	/
	16:46	MeCN (0.2 mL) was added from the syringe	17 measurments collected before addition	/
	16:55	MeCN (0.3 mL) was added from the syringe	20 measurments collected before addition	/
	17:05	MeCN (0.5 mL) was added from the syringe	23 measurments collected before addition lighter in clour, no other change	2 after addition of 2 mL MeCN.jpg
	17:13	The measurment was paused	26 measurments collected before addition	/
	17:14	Flask 1 was cleaned and dried	/	/
		The following steps were done in air:	/	/
	17:15	Flask 2 was removed from the probe and replaced with flask 1	/	/
	17:16	To flask 2 MeCN (HPLC grade, 0.8 mL) was added	/	/
	17:17	The measurment was resumed	/	/
	17:23	/	28 measurments collected before during working aroung the experiment it is likely, that the porbe was touched, which could lead to shift in signal	/
	17:28	The measurment was stopped	30 measurments collected before stopping	/
		The probe was cleaned	/	/

	17:31	The background measurement was performed in air (128 scans)	/	/

Analysis

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
30.05	14:37	AE-503-1	in-situ IR	Mettler Toledo ReactIR 701L with Probe DST 6.35 x 1.5m x 305 DiComp	water	AE-503-1.csv	/	/	No signals observed.
30.05	15:44	AE-503-2	in-situ IR	Mettler Toledo ReactIR 701L with Probe DST 6.35 x 1.5m x 305 DiComp	water	AE-503-2.csv	AE-503-2.PNG AE-503-2_Report.docx	/	For full analysis and interpretation see table below

AE-503-2 Data processing

Step	Descriprtion	Used script	Interpreation	Obtained Plots/Data
1. 3D plotting	Plotting of data in 3D	Test_IR.py	Signals of MeCN seen, sharp peaks in stronger signals	AE-503-2_all_data.png
2. Baseline correction	Baseline correction by substracting the average of measurements in pure MeCN	older version of Test_IR1.py	worked very well, looks like no significant artefacts occure, sharp peaks remain problem	AE-503-2_all_data_after_baseline_correction.png
3. Smoothing	Smoothing of data, trying to get rid of sharp peaks, also last 3 data points in time were omitted, since not usefull for interpreation	older version of Test_IR1.py	Did not really worked, peaks remain	AE-503-2_usefull_data_smoothed.png
4. Peak picking	Omitting smoothing, peak picking by taken 10 most significant peaks	Test_IR1.py	Worked okish, some wired peaks identified, but most relevant peaks were found	AE-503-2_peak_picking.png picked_ir_peaks.csv
5. 2D plot of selected peaks	Plotting data of selected peak, proccesed from origninal data, with Baseline correction and omitting of last 3 data point in the script	Test_IR2.py	Worked good, clear plots obtained, weak signals (max. around 0.015)	selected_ir_bands.csv AE-503-2_2D_identified_signals.png AE-503-2_2D_significatly_shifting_signals_over_time.png AE-503-2_signals_for_further_evaluation.png
6. Calculation of [Ru] concentration	/	AE-503-2_concentration_calculation.xlsx	Worked	/
7. Plotted of conc v. absorption	Plotting of data against conc., c is put in manualy and the measurments at one c are averaged and error bars are shown	Test_IR4.py	Worked, small errorbars different behaviour for different peaks, details see below	ir_vs_concentration.csv AE-503-2_IR_against_conc_all.png AE-503-2_IR_against_conc_error_bars.png
8. Normilization of selceted peaks	Normilization to highest signal	Test_IR3.py	Worked	AE-503-2_normilized.png

AE-503-2 Peaks interpretation

Peak [cm^-1]	general trend with increasing conc.	Further annotation
668	slight decrease	weakly stepwise, close to MeCN signals, linear decrease with c increase
686	slight decrease	weakly stepwise, close to MeCN signals, linear decrease with c increase
696	slight decrease	weakly stepwise, close to MeCN signals, linear decrease with c increase
1246	constant	could be from MeCN
1508	strong decrease	linear with time, not stepwise --> likely artefact, since strong MeCN signal there
1610	strong increase	weakly stepwise, no MeCN signals close to it, strong increase from 0.0075 to approx. 0.015 with increased c
1628	constant	/
1634	constant	/
1644	constant	very slight decrease with increased c
1908	slight increase	would fit to CO region, weakly stepwise, no MeCN signals close to it, slight increase from 0.0035 to 0.0065 increased c
1918	strong increase	would fit to CO region, weakly stepwise, no MeCN signals close to it, strong increase from 0.005 to 0.011 increased c

AE-503-2: better data analysis

Step	Descirption	Used script	Interpreation	Obtained Plots/Data
1. Baseline correction	Baseline correction by taking average of first 5 and last spectra and substract it from the rest, no data cleaned in this step	in_situ_IR_cleaning_and_baseline_AE-503.py	Worked well, MeCN signals are gone, some signals of [Ru] remain	AE-503-2_cleaned_baseline_corrected.csv AE-503-2_baseline_correction.png
2. Concentration calculation	determine c in each dilution step	AE-503-conc.xlsx	Worked, rande from 12 to 2 mM	ru_concentration_AE-503-2.csv
3. Conc plotting	Convert spectra from time domain to conc domain	in_situ_IR_concentration.py	Worked, exp at 0 mM are outlayers	AE-503-2_avg_per_condition.csv AE-503-2_conc.png
4. Baseline correction and Normilization	a. Remove noisy data (both with 0 mM) b. Baseline correction using assymetric least square fit for each spectra c. Normalize spectra to 1182 wavenumbers	in_situ_IR_baseline_correction_and_normilization_AE-503.py	Worked, wavey baseline, singal at 1600 cm-1: at lower c shift to higher wavenumbers and emerging of douplet pattern signal at 1920 cm-1: decrease in intensity, maybe emergingn of new signals at 1940 and 1960 and 1970 cm-1 but could also be noise	AE-503-2_normalized_corrected.csv AE-503-2_normalized_to_1182_zoom_1600.png AE-503-2_normalized_to_1182_zoom_1920.png AE-503-2_normalized_to_1182_all.png

Interpretation

Singals at 1600 and 1920 observed, in 1600 shift of signal observed, wavey behaviour makes further interpreation difficult

Results

Decreasing signals (668, 686, 696) show identical behaviour when normalized, only 1644 show significant lower dectrease and also no change at higher concentrations

Increasing signals (1610, 1908, 1918) show identical behaviour when normalized

Signals below 700 likely artefacts from MeCN or end of range.

Signals at 1610, 1908 and 1918 likely main species of [Ru] --> since two signals in CO area maybe dimer species, but since strongest signals it would hint to main species, not to small amount in equilibrium. Also 1908 and 1918 could be part of same signal

Signal 1644 --> no real idea, would fit to monomer, but to small signal, no signal matching signal for CO found

In general: we see signals, but nothing which can obviously be attributed to monomer and dimer

Linked experiments

- - AE-152: Degassing of MeCN
- Prep work - AE-442: Degassing of Milli-Q water
- Two-photon - AE-456: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h
- Two-photon - AE-473: Testing of in-situ IR probe
- Two-photon - AE-493: Dilution of [Ru(CO)(OH)2(PNP)] solution in water while in-situ IR measurement
- Two-photon - AE-499: Stocksolution of [Ru(CO)(OH)2(PNP{tBu})] in water

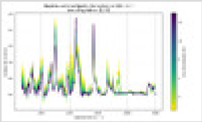
Linked resources

- Equipment - ReactIR 701L with DST 6.35 x 1.5 m x 305 DiComp probe
- Protocol - In-situ IR measurment with ReactIR 701L in GL14 flask under inert conditions

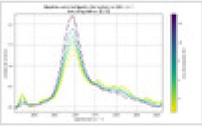
Attached files

AE-503-2_normalized_to_1182_all.png

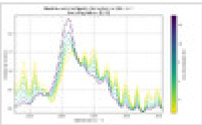
sha256: 235521e934447467d5a91b6f902f4ddce2ce7e2a67fde3586b6e83f770556b70



AE-503-2_normalized_to_1182_zoom_1920.png
sha256: 6979edcc5d6ce4e493256ea54fe1647dd000c0a7cb09bd84a5a48578683c96fe



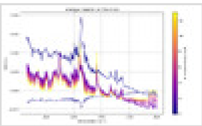
AE-503-2_normalized_to_1182_zoom_1600.png
sha256: 7be822a612bef03c26e3a17f58ac89fec513f0db27b559cf553ea65bfccc48d7



AE-503-2_normalized_corrected.csv
sha256: 6c1732a03738c566191660e5b9e24e6ae14f2445e4e396e3ecc88c21b8d37c12

in_situ_IR_baseline_correction_and_normilization_AE-503.py
sha256: 951f3e4d4349cd9c02ad74a81ee976cd6ff1423fe001a704858b7cea3cd95393

AE-503-2_conc.png
sha256: a6fa5fe30a8564d9f9be43177b171c5ef5f69c79c50767fa2ca6552b25b730c1



AE-503-2_avg_per_condition.csv
sha256: bb9b121f405f49efa5120dac00c44052e01bf80aaea097d29047da80c2761b73

in_situ_IR_concentration.py
sha256: 8b0627a7b06b63e3032ea2d009a1b4b69cbfd36e58761bedad0dc3e57e5b849b

ru_concentration_AE-503-2.csv
sha256: fc6fcde7843828ace8cfed6edd58ec3e7cc366e5dfc9f32ee64b556e7f615ab0

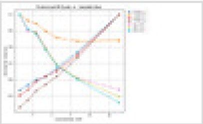
AE-503-conc.xlsx
sha256: 87d6796aeb8bc2855a9d80caf35d02db9064f8ded7515a6ea8be62e19af5a1df

AE-503-2_baseline_correction.png
sha256: c2c348f51e75c766f32357fbb1306aa9e95b97b79836c1e51eea8c5588ede62a

AE-503-2_cleaned_baseline_corrected.csv
sha256: e025cd203ecb180c2ba01406026e85df8bb391e99041d88f9317d61f7e7f9736

in_situ_IR_cleaning_and_baseline_AE-503.py
sha256: 2828c08894c2af27baca537f6ea03231c2acd91bcbf53854dab4ac53d26ce30e

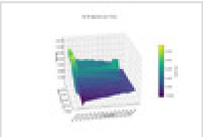
AE-503-2_normilized.png
sha256: a867efbbcac7b23524f04a6aa9aff8e007626a9cdf27980439b6de4076edafb5



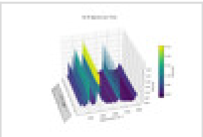
Test_IR3.py
sha256: b6de95cd4302f8ce24f685ee8e60704cd34ba04248d73b52b4e628c330ec32a6

Test_IR.py
sha256: a352997853cac05e8beda71a1aa8e3e1c90098972e9e21660265cec2be87c53a

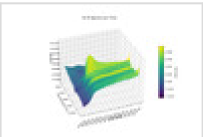
AE-503-2_all_data_after_baseline_correction.png
sha256: 3d72b1d0acc51730b7b5d98147c23a747a828772fe60c6366a8faf55c09227d7



AE-503-2_all_data.png
sha256: 42eff2dd5d050e3097a86264dc3dbd2a255f5ed631e9b12638d98b17a4c1f4ed



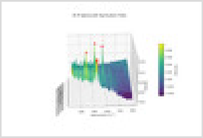
AE-503-2_usefull_data_smoothed.png
sha256: 81e03a847877d29a9594307331082ce3c83a00976d0b98de11fbd7267d5cbddc



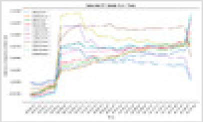
Test_IR1.py
sha256: 32543411845b6b15f4d1cd70341286962e26b40a4025dca26bddf1b6da417133

picked_ir_peaks.csv
sha256: bbc8dd9dbe9e90ebe0259e444915016978ed9a0289be0ff5a86810bf4dad7c6e

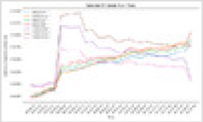
AE-503-2_peak_picking.png
sha256: 2382bc52f9e5cf9d331963225eec24a91e81a73f84329003ad9a2ee5361ef656



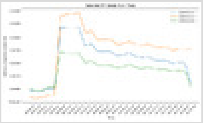
AE-503-2_2D_identified_signals.png
sha256: 74d0dd10e5f5935160dfc93311b5de8330dede33ca5417314366ea7f39cd3fe0



AE-503-2_2D_significatly_shifting_signals_over_time.png
sha256: e005bc0846c4c04d44bfc313a6c8535a251a1fa5d837a5c1fe904a7d61fb6354



AE-503-2_signals_for_further_evaluation.png
sha256: c7a4dc6029c0aaa867ab0da04858ea0d05145c71397fca3bfd68bbfd9db1b289

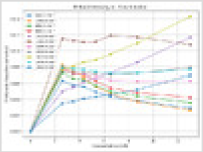


Test_IR2.py
sha256: 724c7e2654ee92fec4b06ec3defbe0131ba619b5f681d721ff303e5b6aaf9497

selected_ir_bands.csv
sha256: 283b1ce4dd704caf8b54265a710faf02177d4721fe233bff438293b2c77813d3

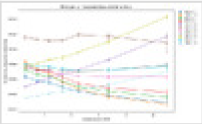
AE-503-2_concentration_calculation.xlsx
sha256: c2a9fae0d38958deba6dfb0892f02812a20ecea2df3aebe76009b3f021820807

AE-503-2_IR_against_conc_all.png
sha256: 4c8c34ec8d0e8c6db572c2c84c064d0942a9cf25bf67af83a720d0372c4a7e2e



Test_IR4.py
sha256: 89dbd587a72b9658a5f1ef9f8a0e9c91da10690a8a4d45b8d5ad59bdec632a65

AE-503-2_IR_against_conc_error_bars.png
sha256: 7167686805d7255f6c8d73d433f7c02969d9ca783b49ea4855299b4daf4fc41b

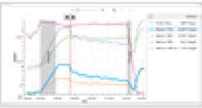


ir_vs_concentration.csv
sha256: 6f294d248a38552be347b2dc734a3d4f2806d90ce538cc26f7f2bb7d351c992d

AE-503-1.csv
sha256: 30ae6c8d3a39f38a7f7a5e6e1287de93656e5226913ddc7055f1ec7dbf67d385

AE-503-2.csv
sha256: cb6e6b309f4b86f3e12de91ce60fb622464c3fb85930a8b9c6d2e06fcf7f07af

AE-503-2.PNG
sha256: b9eb886ca5012a9a91eaadf640324f70caa85538a818014845fbd89a8f0dc57f



AE-503-2_Report.docx
sha256: 9eb410cbfff076c4d915faf1dcb3bffbe95fdd3d7061e731bc4ecfc0ccf18519

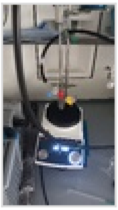
2-after-addition-of-2-mL-MeCN.jpg
sha256: 61da293f994d6e402d30fc7a4a0b89af1e94d8e02939e52a308725a2d1aeed28

2-after-addition-of-1-mL-MeCN.jpg
sha256: 3b8d08c464c825490720c72eeb17b7b092ccd11d38c83d40a35d5afba22f700f

2-after-addition-of-0.6-mL-MeCN.jpg
sha256: 91e6c1943676eb92ca4b0ab741df67607aaa0aae9b3f1f86cca2002f43ca23e8

2-after-addition-of-0.4-mL-MeCN.jpg
sha256: 1c1faeb372322bd89f1f1c75faf04eaaf60e066d76fdc5355bf2128ce5592ee0

2-setup-during-sekurating.jpg
sha256: 909eb1d6631a2a65db98be80e075eab4cccaa887cca781886a377566fffac0e7



2-setup-flask-1.jpg
sha256: 7bb87bd11a2cfa1e15e742066a1f7b696c51acc55b4822ec77553bdc67774616

2-positon-of-probe-in-flask-2.jpg
sha256: 02f00a638e2930e77b94b37ab82b3f05dc1ca2d9d7942e85edad104142acbd57

1-setup-flask-2.jpg
sha256: 76af5b093263f49ee1c739252b2accb597744cfd4c19875c2d5056dffa7e2741

1-position-of-probe-in-flask-2.jpg
sha256: c5ad662dbff23777aadcf82edd8dd28a9ba220c62ab7af8ab3a1999c950aa7e2

1-setup-flask-1.jpg
sha256: a3089198ff30204c57ee8bb623f4aabb521632d13d74a1900187eaa01419266d

1-position-of-probe-in-flask-1.jpg
sha256: 6721285237766f2453943ecea34c9feccacdf114215c717a4cc46d34237c0e66

AE-493.cdxml
sha256: df124f38dd8ca112a9ed9e47dc25758df4cf442a4bb657f1821b70595f507791

Unique eLabID: 20250530-774c50cab8725d5d010b2ab937391ce030940c79
Link: https://elab.water-splitting.org/experiments.php?mode=view&id=2213